

Brinta Kundu

Software Engineer | Full-Stack & Machine Learning

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About

Computer Science **graduate** from Hunter College with experience building **AI-powered applications, full-stack web platforms**, and **iOS applications**. Experienced with **React, FastAPI, PostgreSQL, Swift, and Firebase**, with a passion for developing software that improves accessibility, education, and community wellbeing. Strong collaborator who enjoys solving challenging engineering problems and learning new technologies.

Skills

Programming Languages

Python, TypeScript, JavaScript, Swift, SQL, C++, HTML/CSS, R

Frameworks & Technologies

React, Next.js, FastAPI, PostgreSQL, Firebase, REST APIs, Docker, Git, pgvector, SQLAlchemy, Alembic

Professional Skills

Software Development, Team Collaboration, Problem Solving, Agile Development, Technical Documentation, Debugging

Education

Hunter College

Bachelor of Science in Computer Science

GPA: 3.8

August 2023 – May 2026

Bard High School Early College

Associate of Arts & High School Diploma

September 2019 – June 2023

Professional Experience

Software Engineer Intern

Trance4mation Games

Jan 2026 – May 2026

- Contributed to the development of **interactive digital experiences** focused on emotional wellness and community engagement using **React, Next.js**, and **TypeScript**.

- Collaborated with engineers and designers during **feature planning**, implementation, and testing to deliver production-ready improvements.
- Participated in **code reviews**, debugging, and quality assurance to improve application reliability and maintainability.
- Worked within an **Agile development environment**, communicating progress and incorporating stakeholder feedback into product iterations.

iOS Frontend Developer Intern

Stretch Mode (formerly Movement Vault)

Aug 2025 – Dec 2025

- Developed and enhanced **Swift-based iOS user interfaces**, creating clean and maintainable frontend components.
- Diagnosed and resolved UI issues through debugging and testing, improving application stability and user experience.
- Utilized **Git** for version control while collaborating with teammates throughout the software development lifecycle.

Tutor

Aug 2023 – Present

- Provide individualized instruction to **K–12 students** across mathematics, science, and computer science subjects.
- Design personalized learning strategies that strengthen **problem-solving**, analytical thinking, and independent learning.
- Adapt communication styles to students with varying learning needs, improving engagement and academic performance.

Projects

IntelliDoc

AI-Powered Document Management Platform

- Developed an **AI-powered document management system** enabling semantic search across uploaded files.
- Built backend services using **FastAPI**, **PostgreSQL**, **SQLAlchemy**, and **pgvector** to support efficient document retrieval.
- Implemented **BGE-M3 embeddings** for vector search, significantly improving document relevance over traditional keyword search.
- Designed REST APIs for authentication, document uploads, folder organization, and file management in a **four-person Agile team**.
- Integrated secure authentication and scalable database architecture to support future platform growth.

Warehouse DQN Robot Navigation

Deep Reinforcement Learning | PyTorch

- Developed a Deep Q-Network (DQN) agent from scratch using **PyTorch** to autonomously navigate randomly generated warehouse environments.
- Designed a custom grid-world simulation with procedural obstacle generation and **BFS-based validation** to guarantee solvable training environments.
- Implemented experience replay, **target networks**, **epsilon-greedy exploration**, and reward shaping to improve **learning stability** and convergence.
- Built a training pipeline with performance metrics and visualizations to monitor reward, loss, exploration rate, and success rate across **thousands of training episodes**.

Interests

Reading, Game Development, Machine Learning, Bioinformatics